ON-THE-JOB TRAINING REPORT

STUDENTS MARKS PREDICTION

A Machine Learning Based Academic Performance Prediction Project

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SUBMITTED TO

SHIVAJI UNIVERSITY, KOLHAPUR

For partial fulfilment of the degree of

B.Sc. Computer Science (Entire) – Part III

SUBMITTED BY

SANIKA SUKUMAR KOLE

Department of Computer Science
Jaysingpur College, Jaysingpur
Academic Year: 2026–2027

PRN No.: 2024011135
Roll No.: 35

CERTIFICATE

This is to certify that the On-the-Job Training report entitled “Students Marks Prediction” is submitted by Sanika Sukumar Kole for partial fulfilment of the B.Sc. Computer Science (Entire) degree requirements of Shivaji University, Kolhapur. The report presents an academic project design and documentation concerning the use of student academic and study-related features to estimate final marks using Machine Learning.

The report is submitted for review under the guidance and supervision of the concerned faculty. Details such as PRN number, guide names, signatures, and dates are to be completed and verified by the college.

Place: Jaysingpur

Date: _____

Student	Faculty Guide	Co-ordinator
Sanika Sukumar Kole	_____	_____

DECLARATION

I hereby declare that this report entitled “Students Marks Prediction” has been prepared by me as an academic project report for the B.Sc. Computer Science (Entire) programme. The report explains the proposed problem, dataset structure, machine learning workflow, interface design, testing approach, limitations, and future scope of a student marks prediction system.

I understand that any screenshots labelled as demonstration mockups and any sample values included in this report are illustrative. They must not be interpreted as evidence of a live deployment, actual student results, or measured model performance. Any institutional or training details must be verified and completed before final submission.

Place: Jaysingpur

Date: _____

Name of Student: Sanika Sukumar Kole

PRN No.: 2024011135

Signature of Student: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Shivaji University, Kolhapur, and Jaysingpur College, Jaysingpur, for providing the academic environment and opportunity to learn about practical applications of Computer Science.

I am thankful to my faculty guide, Head of the Department, Principal, and all teaching staff for their guidance, encouragement, and support during the preparation of this project report. Their academic direction helps me connect classroom concepts with practical problem-solving.

I would also like to thank my family and friends for their motivation and support. This project has helped me understand how data preparation, statistical learning, and software interfaces can be combined to create a useful academic support tool.

Name of Student: Sanika Sukumar Kole

Class: B.Sc. Computer Science (Entire) – Part III

Academic Year: 2026–2027

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1. STUDENT INFORMATION

Particulars	Details
Name of Student	Sanika Sukumar Kole
Class	B.Sc. Computer Science (Entire) – Part III
PRN Roll No.	2024011135 35
Project Title	Students Marks Prediction
Institution	Jaysingpur College, Jaysingpur
University	Shivaji University, Kolhapur
Academic Year	2026–2027
Project Domain	Machine Learning / Educational Data Analysis
Faculty Guide	To be confirmed by college
Training Organisation	Not provided

Project scope

The report describes a prototype that estimates a student's final marks from selected academic and study-related inputs. It is intended for learning and demonstration. A prediction is an estimate, not a guarantee of a student's actual result, and it should not be used to label or limit a student's potential.

2. INTRODUCTION

2.1 Background

Educational institutions collect different kinds of academic information, such as attendance, internal assessment marks, assignment scores, previous examination results, and study habits. Reviewing these factors together can help teachers and students understand patterns related to academic performance.

Machine Learning is a branch of Artificial Intelligence that allows a computer to learn patterns from examples and use them to make predictions. In a student marks prediction project, a model can be trained using historical records where input features and final marks are available. Once trained and evaluated, it can estimate a likely final score for a new set of input values.

2.2 About the Project

The project titled “Students Marks Prediction” proposes a simple, user-friendly application that takes student-related academic inputs and predicts an estimated final mark out of 100. The prototype uses a supervised regression approach, with Linear Regression presented as the baseline algorithm. Inputs may include study hours, attendance percentage, previous exam marks, assignment average, and internal test marks.

The system is intended as a learning aid. It does not replace teachers, formal assessment, or a student's own effort. Reliable performance depends on the quality, size, representativeness, and privacy protection of the dataset used to train it.

2.3 Problem Statement

To design and document a Machine Learning prototype that uses selected academic and study-related features to estimate final student marks, and presents the result in an understandable form.

2.4 Proposed Solution

Prepare a structured dataset, clean and analyse the data, separate it into training and testing sets, train a regression model, evaluate its errors, and connect the model to a simple interface for entering values and viewing an estimated score.

3. OBJECTIVES OF THE PROJECT

• To understand the fundamentals of supervised Machine Learning and regression.

• To identify meaningful academic features that may be associated with final marks.

• To prepare a structured dataset suitable for model training and testing.

• To apply data cleaning and basic exploratory data analysis using Python.

• To train a baseline Linear Regression model for estimating final marks.

• To evaluate prediction quality using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

• To design a simple interface for accepting input values and displaying a predicted score.

• To improve coding, debugging, documentation, analytical thinking, and project presentation skills.

• To understand ethical considerations such as student privacy, fairness, and the limitations of predictions.

4. PROJECT OVERVIEW

4.1 System Description

The proposed application follows a simple workflow. The user enters study and academic details. The input values are checked and converted into the format expected by the model. The trained model then generates a numeric estimate of final marks. The interface displays the estimate with a short note explaining that it is not an official result.

4.2 Main Modules

Module	Description
Data collection	Organises historical academic records and relevant features.
Data preprocessing	Checks missing values, data types, ranges, and duplicate records.
Exploratory analysis	Reviews distributions, relationships, and possible outliers.
Model training	Fits a regression model using training records.
Model evaluation	Measures error and goodness of fit on held-out records.
Prediction interface	Collects new input values and displays the estimated marks.
Documentation	Records design choices, screenshots, testing, limitations, and references.

4.3 Expected Outcome

The expected outcome is a working academic prototype that demonstrates the end-to-end process of a Machine Learning prediction task. Actual predictive quality can only be reported after training and testing the model on a suitable dataset; no verified accuracy or error score is claimed in this report.

5. TECHNOLOGIES AND TOOLS USED

Technology / Tool	Purpose
Python	Main programming language for data processing and model development.
Pandas	Loading, cleaning, filtering, and analysing tabular data.
NumPy	Numerical operations and array handling.
Scikit-learn	Train-test split, Linear Regression, and evaluation metrics.
Matplotlib / Seaborn	Visualising distributions and relationships in the dataset.
Streamlit (proposed UI)	Creating a simple interactive web interface for prediction.
CSV	Simple format for storing structured example records.
VS Code / Jupyter Notebook	Writing code, running experiments, and debugging.

5.1 Why Linear Regression?

Linear Regression is a common baseline algorithm for predicting a continuous numeric value. It learns a relationship between input features and a target value. In this project, the target is final marks. It is relatively easy to explain and provides a useful starting point before comparing more advanced models.

5.2 Software and Hardware Requirements

Software: Python 3, a code editor or notebook, Pandas, NumPy, Scikit-learn, Matplotlib, and optionally Streamlit. Hardware: a standard laptop or desktop capable of running Python. A small educational dataset does not normally require specialised hardware.

6. TASK UNDERTAKEN

The main task is to plan a student marks prediction workflow and document its components. The practical implementation should follow these stages:

• Define the prediction target and choose input features that are available before the final examination.

• Create or obtain an authorised dataset, ensuring that personal identifiers are removed or protected.

• Inspect the dataset for missing values, invalid percentages, inconsistent data types, and duplicates.

• Perform exploratory data analysis to understand feature distributions and correlations.

• Split records into training and testing subsets before model evaluation.

• Train a Linear Regression baseline using the training subset.

• Generate predictions on held-out test records and calculate MAE, RMSE, and R^2.

• Build a form-based interface and validate the ranges of the entered values.

• Test ordinary, boundary, and incomplete input cases and document observed results.

• Prepare screenshots, report sections, and references for academic submission.

Expected Deliverable

A documented project prototype consisting of a dataset, preprocessing and training workflow, evaluation output, a prediction interface, and this report. Screenshots in the next section are visual demonstration mockups created for the report; they are not screenshots captured from a running application.

7. RESULTS AND SCREENSHOTS

The following images illustrate the proposed application layout and workflow. They are demonstration mockups prepared for report presentation. Replace them with genuine screenshots after implementing and running the application if your college requires proof of execution.

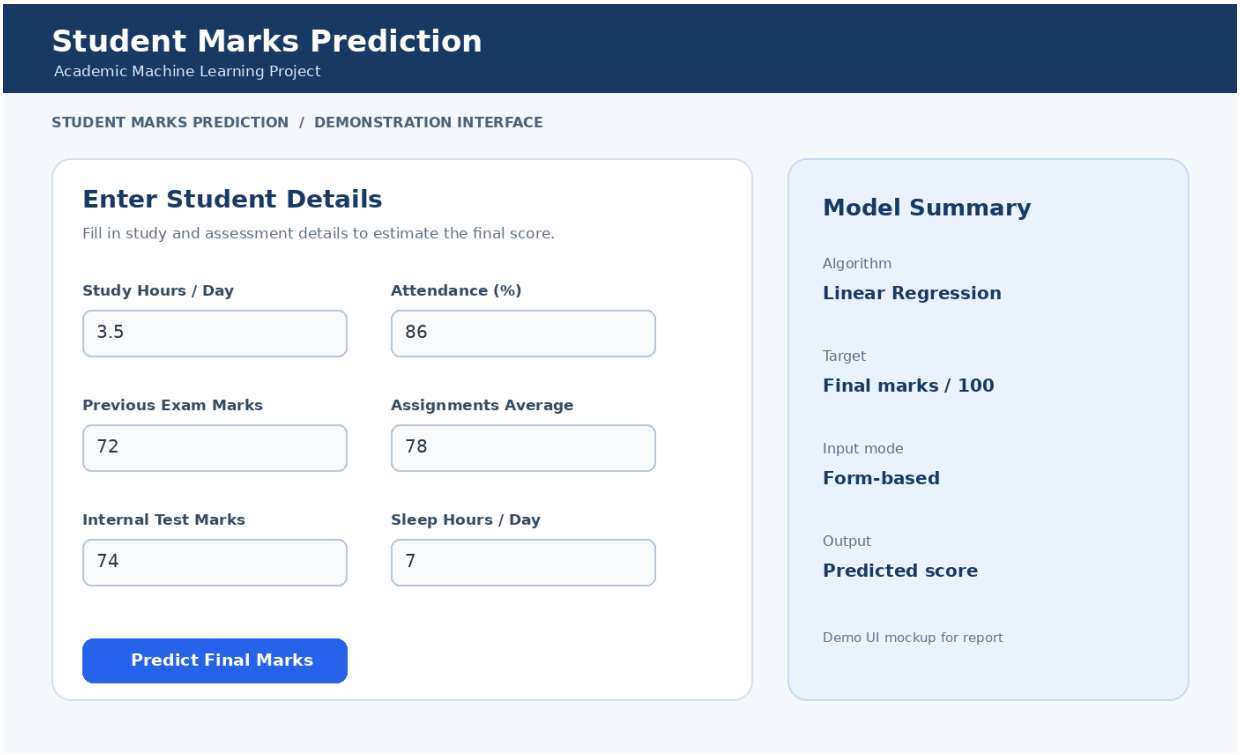


Figure 1: Proposed home page and student input form (demonstration mockup).

The screen shows example fields for study hours, attendance, previous examination marks, assignment average, internal test marks, and sleep hours. In a real implementation, field names should match the features actually used to train the model.

7. RESULTS AND SCREENSHOTS (CONTINUED)

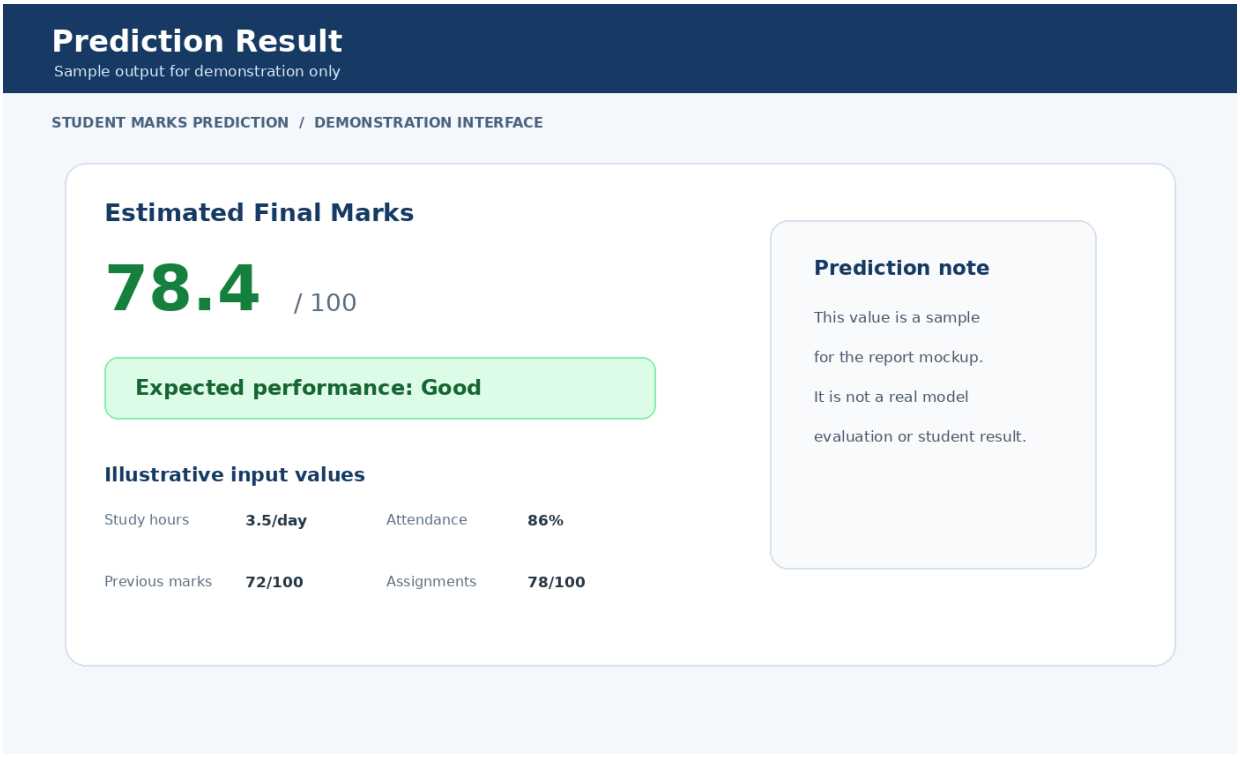


Figure 2: Example prediction result screen (demonstration mockup).

The score shown in this image is illustrative only. It is not produced by a trained model and must not be reported as a real prediction or performance result. A real application should display the model output, along with a clear statement that it is an estimate.

7. RESULTS AND SCREENSHOTS (CONTINUED)

Dataset Preview

Example structured student records

STUDENT MARKS PREDICTION / DEMONSTRATION INTERFACE

Study hrs	Attendance	Previous	Assignment	Internal	Final marks
3.5	86	72	78	74	79
2.0	70	58	62	60	63
5.0	94	88	91	90	92
1.5	65	49	55	52	54
4.0	82	76	80	78	81
2.8	75	66	70	68	70
6.0	96	92	95	94	96

Example rows only — use an actual dataset for genuine training and evaluation.

Figure 3: Example dataset preview (demonstration mockup).

The displayed rows are sample values designed to demonstrate the dataset structure. For genuine training, use records with a consistent definition of the final-marks target and ensure that the data is authorised, relevant, and sufficiently representative.

7. RESULTS AND SCREENSHOTS (CONTINUED)

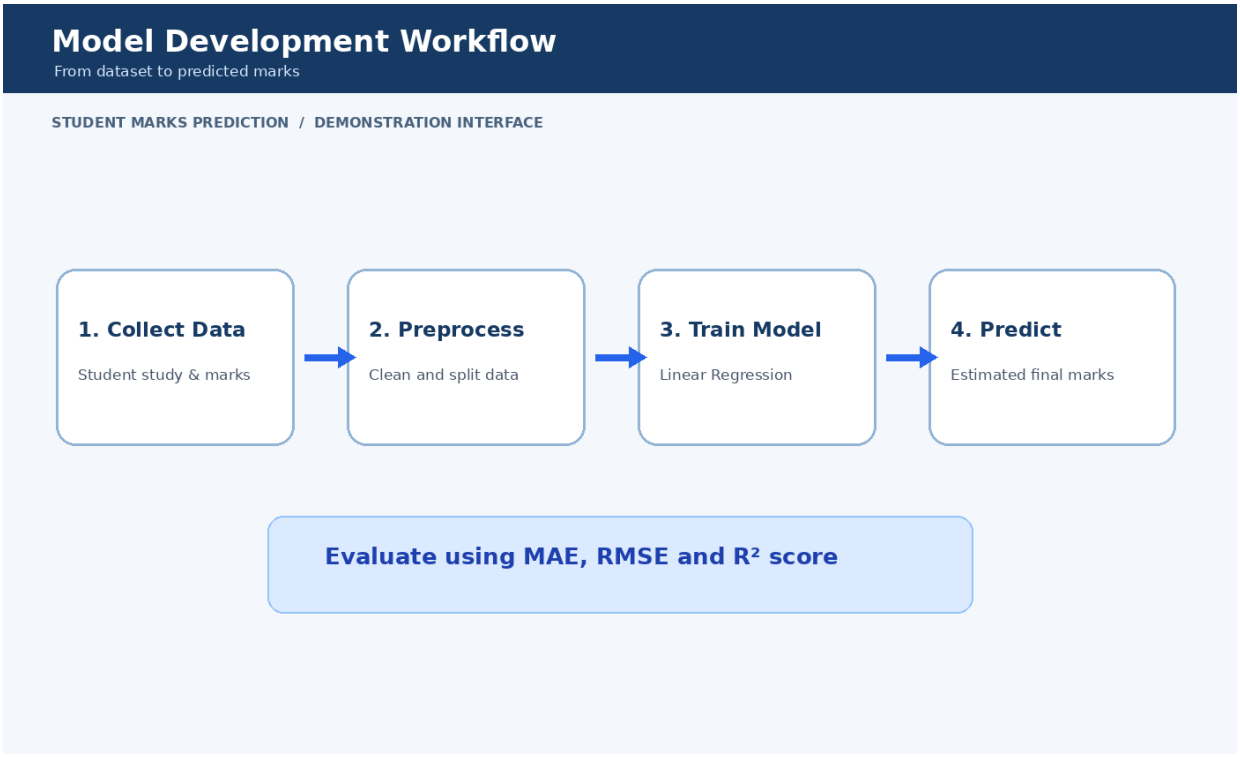


Figure 4: Proposed model development workflow (demonstration mockup).

The workflow begins with data collection, continues through preprocessing and model training, and ends with prediction and evaluation. Evaluation should be based on held-out records, not on the training data alone.

8. SKILLS AND KNOWLEDGE ACQUIRED

• Python programming: using variables, functions, libraries, and scripts for a data workflow.

• Data handling: loading CSV files, checking missing values, selecting columns, and validating ranges.

• Machine Learning: understanding supervised learning and the purpose of regression.

• Model evaluation: interpreting MAE, RMSE, and R^2 without confusing them with classification accuracy.

• Interface design: arranging input fields and presenting numeric outputs clearly.

• Testing and debugging: tracing errors, checking invalid inputs, and validating output behaviour.

• Documentation: describing the problem, methodology, screenshots, limitations, and references.

• Responsible data use: protecting student information and avoiding overconfident interpretation of predicted marks.

9. LEARNING EXPERIENCE

This project provides an opportunity to connect programming concepts with a practical educational use case. It demonstrates that a prediction system is not just a model: it also requires a well-defined problem, trustworthy data, careful preprocessing, meaningful evaluation, a usable interface, and clear communication of limitations.

A key learning point is that a model can produce a number even when the data is weak. Therefore, results must be checked rather than accepted automatically. The test dataset should be separate from the training dataset, and the report should present measured evaluation values only after the experiment has actually been run.

The project also highlights the importance of student privacy. Academic records should be used only with proper permission, unnecessary identifying details should be excluded, and predictions should support learning rather than discourage students or determine opportunities unfairly.

The screenshots in this report are mockups to explain the expected interface. Capturing genuine screenshots after running the project is an important final documentation step.

10. PROJECT DEVELOPMENT METHODOLOGY

10.1 Requirement Analysis

Identify the target value (final marks out of 100), the users of the application, and the features that can reasonably be known before the final examination.

10.2 Dataset Preparation

Collect authorised historical records or create a clearly labelled demonstration dataset. Keep the target and feature definitions consistent. Remove direct identifiers and document the source and limitations of the data.

10.3 Data Preprocessing

Inspect null values, duplicate rows, data types, and impossible values. For example, attendance should be within 0–100 percent and marks should follow the grading scale used by the institution. Decide how to handle missing data before model fitting.

10.4 Train-Test Split

Separate the dataset into training and testing subsets. The model learns from the training subset; the test subset is reserved to estimate how well it generalises to unseen records. If records are grouped by class, year, or student, consider a split that avoids leakage across related records.

10.5 Model Training

Fit a baseline Linear Regression model using the selected input features and the final-marks target. Any transformations or preprocessing steps should be fitted on the training data only and then applied consistently to test data.

10.6 Evaluation and Interface

Evaluate the held-out predictions with MAE, RMSE, and R^2. Build the interface only after the input schema and model pipeline are stable. Validate input ranges and display the prediction with an explanatory note.

11. DATASET DESCRIPTION

The dataset is the foundation of the prediction system. The exact dataset source and record count have not been supplied, so this report does not claim that a real dataset has already been collected or that a model has been trained.

Property	Description
Dataset name	student_marks_data.csv (suggested filename)
Data type	Structured tabular data
Source	To be documented after dataset is selected
Number of records	To be confirmed after dataset preparation
Target column	final_marks
Example input columns	study_hours, attendance, previous_marks, assignment_average, internal_test_marks
Data privacy	Exclude names, phone numbers, addresses, and other unnecessary identifiers.

11.1 Sample Schema

Column	Type	Example / allowed range
study_hours	Numeric	0–16 hours/day (validate contextually)
attendance	Numeric	0–100 percent
previous_marks	Numeric	0–100, depending on assessment scale
assignment_average	Numeric	0–100, depending on assessment scale
internal_test_marks	Numeric	0–100, depending on assessment scale
final_marks	Numeric target	0–100, depending on assessment scale

11.2 Data Quality

Before training, check for missing values, repeated records, unusual outliers, and mismatched assessment scales. Do not include features that would only become known after the final marks are awarded, because that would leak information into the model.

12. FEATURE DESCRIPTION

Sr.	Feature	Description / role
1	Study hours per day	Approximate daily study time; self-reported and potentially noisy.
2	Attendance (%)	Percentage of attended classes, measured consistently.
3	Previous exam marks	Earlier assessment result that may provide academic context.
4	Assignment average	Average score across relevant assignments.
5	Internal test marks	Internal assessment score available before the target result.
6	Final marks (target)	The numeric outcome the model is trained to estimate.

The final feature list should be determined by the available dataset and educational context. Adding more features does not automatically improve model quality. Features should be selected because they are relevant, available at prediction time, and ethically appropriate.

12.1 Target Variable

The target variable is the student's final marks, represented as a numeric value. This makes the task a regression problem rather than a classification problem. If the project later predicts categories such as “Needs Support” or “On Track”, that would be a separate classification task and would require a clearly defined label rule.

12.2 Ethical Use of Features

Avoid sensitive or irrelevant personal characteristics. A student's predicted score should not be used as a fixed judgement of ability. Students learn at different rates, and any automated estimate should be interpreted alongside teacher feedback, context, and actual assessment evidence.

13. MODEL TRAINING AND EVALUATION

13.1 Model Training

Linear Regression estimates a numeric target as a weighted combination of input features. The model learns coefficients from the training records. Its simplicity makes it a useful baseline, but it may not capture complex nonlinear relationships between study patterns and marks.

13.2 Evaluation Metrics

Metric	Meaning	Interpretation
MAE	Mean Absolute Error	Average absolute difference between predicted and actual marks; lower is better.
RMSE	Root Mean Squared Error	Penalises larger errors more strongly; lower is better.
R^2 score	Coefficient of determination	Measures how much target variation is explained relative to a baseline; closer to 1 is generally better, but negative values are possible.

13.3 Results

No verified evaluation values are reported in this document because a real dataset and executed training results were not supplied. After running the project, enter the measured MAE, RMSE, and R^2 values below. Do not substitute illustrative numbers or claim “accuracy” for a regression model.

Evaluation item	Actual measured value
MAE	_____
RMSE	_____
R^2 score	_____
Test records used	_____

13.4 Avoiding Data Leakage

All steps that learn from data, including imputation and scaling where used, should be fitted only on the training subset. Test records must not influence model selection or preprocessing decisions. This helps make the evaluation more realistic.

14. TESTING AND VALIDATION

Test ID	Scenario	Expected verification
TC-01	Typical valid inputs	The form accepts values within allowed ranges.
TC-02	Attendance = 0 or 100	Boundary values are accepted or handled according to validation rules.
TC-03	Marks outside 0–100	The application rejects or clearly flags invalid values.
TC-04	Missing required field	A helpful validation message is displayed.
TC-05	Negative study hours	Invalid values are rejected.
TC-06	Prediction button	The model is called and a numeric estimate is shown.
TC-07	Model file unavailable	The application displays a useful error instead of crashing silently.
TC-08	Repeated prediction	The interface remains responsive and produces consistent results for identical inputs.

14.1 Validation Approach

Validation should include normal values, boundary values, invalid values, and incomplete inputs. The model should also be checked on unseen records. Record actual pass/fail outcomes after executing each test; the table above describes test cases, not completed test evidence.

14.2 User Interface Checks

Check that labels are understandable, input ranges are shown, errors are readable, the result is visible, and the page explains that the prediction is only an estimate. Avoid displaying unnecessary student identifiers.

15. LIMITATIONS AND FUTURE SCOPE

15.1 Limitations

• Prediction quality depends on the quantity, quality, and representativeness of the dataset.

• Study hours and similar self-reported variables may be inaccurate or inconsistent.

• Linear Regression may not model complex relationships between features and marks.

• School, subject, assessment, and grading differences can make records difficult to compare.

• Predictions may be biased if historical data reflects unequal access to resources or support.

• The current report documents a proposed prototype; it does not claim a verified live deployment or measured performance.

15.2 Future Scope

• Use a larger, authorised, and appropriately anonymised dataset.

• Compare Linear Regression with Random Forest Regressor and other suitable algorithms using the same evaluation split.

• Add charts showing feature distributions and model residuals.

• Develop a Streamlit interface with input validation and downloadable results.

• Add explainability features to show which factors are associated with the estimate, while avoiding causal claims.

• Evaluate fairness and performance across relevant groups only when lawful, ethical, and supported by adequate data.

• Deploy the application securely with access controls and clear privacy notices.

16. CONCLUSION

The Students Marks Prediction project demonstrates how Machine Learning can be applied to an educational data problem. The proposed system accepts selected academic and study-related features and estimates a numeric final mark using a regression model. The report covers the project objectives, proposed architecture, dataset schema, feature descriptions, methodology, evaluation metrics, interface mockups, testing plan, limitations, and future improvements.

The project is valuable as a learning exercise because it brings together Python programming, data preparation, model training, evaluation, and user-interface design. However, a prediction system is only as reliable as the data and validation process behind it. Real evaluation results must be obtained from an actual implementation and a suitable held-out test set before any claim about performance can be made.

In conclusion, the prototype can be developed further into an academic support tool that encourages informed discussion and timely learning support. It should be used responsibly, protect student privacy, and never be treated as a final judgement of a student's ability or future success.

Place: Jaysingpur

Date: _____

Student Signature: _____

17. REFERENCES

The following official documentation provides background for the tools and evaluation methods suggested in this report. The final submitted project should cite any additional dataset, tutorial, or research source actually used.

1. Python Software Foundation. Python Documentation. https://docs.python.org/3/

2. Pandas Development Team. pandas Documentation. https://pandas.pydata.org/docs/

3. Scikit-learn Developers. Linear Models: Ordinary Least Squares; Model Evaluation. https://scikit-learn.org/stable/

4. Streamlit. Streamlit Documentation. https://docs.streamlit.io/

5. NumPy Developers. NumPy Documentation. https://numpy.org/doc/

6. Matplotlib Development Team. Matplotlib Documentation. https://matplotlib.org/stable/

APPENDIX: FINAL SUBMISSION CHECKLIST

• Enter the correct PRN number and confirm the class and academic year.

• Confirm the faculty guide, co-ordinator, principal, training organisation, and training dates with the college.

• Replace demonstration mockups with genuine screenshots from the running application if required.

• Attach the actual dataset description and source/permission details.

• Run the model and enter measured MAE, RMSE, and R^2 values.

• Verify page numbering, signatures, formatting, and college submission requirements.